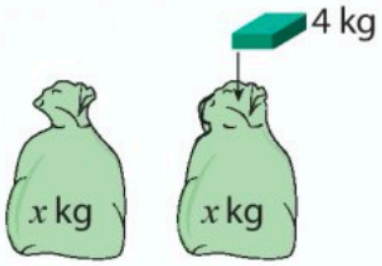
Section 8E Using Equations to Solve Problems

When 4 kg is added to 2 bags of equal weight the total weight is 16 kg. What is the weight of each bag?



Let x kg be the weight of each bag.

Then 2 bags weigh $2(x)$ kg $= 2x$ kg

When 4 kg is added to the 2 bags we get $2x + 4$ kg

Therefore $2x + 4 = 16$

$2x + 4 - 4 = 16 - 4$

$2x = 12$

$x = 6$, each bag must weigh 6 kg.

Example 1

Jennie has some computer games. Mia has 3 games more than Jennie.
Together they have 19 games.
How many games do they each have?

Let n stand for the number of games that Jennie has.
Mia must have $n + 3$ games
Together they have $n + n + 3 = 2n + 3$

$\therefore 2n + 3 = 19$

$2n + 3 - 3 = 19 - 3$

$2n = 16$

$n = 8$

Jennie has 8 computer games and Mia has $8 + 3 = 11$ games

Exercise 8E

1. Sean has some computer games. Roisin has 4 fewer that Sean.
Together they have 16 games.
Let n stand for the number of games Sean has.
(i) Write an expression for the number of games Roisin has.
(ii) Write an expression, in terms of n, for the number of games they have together.
(iii) Use this expression to form an equation in n.
(iv) Solve this equation and write down the number of games each has.

2. Peter buys 3 bars of chocolate costing €x each and a bag of sweets costing €$(x + 4)$.
He spends a total of €12.
(i) Write down an equation in x
(ii) Solve the equation.
(iii) How much did the bag of sweets cost?

3. Dorothy has some cds.
Elaine has 5 more than Dorothy.
Michael has 10 more than Dorothy

Altogether they have 105 cds.
Let n stand for the number of cds Dorothy has.
(i) Write an expression for the number of cds Elaine has.
(ii) Write an expression for the number of cds Michael has.
(iii) Write an expression, in terms of n, for the number of cds all three have together.
(iv) Use this expression to form an equation in n.
(v) Solve this equation and write down the number of cds each has.

4. Tom has two brothers, Colin and Doug
Colin is 4 years younger than Tom
Doug is twice as old as Tom.
Let y stand for Tom's age in years
(i) How old is Colin in terms of y
(ii) How old is Doug in terms of y
(iii) If their ages add up to 24 write an equation in y connecting their ages.
(iv) Solve the equation and find out the age of each brother.

5. The following is a number puzzle
"Think of a number and multiply it by 2 then add 12.
The answer is 3 times the starting number".
What number did I start with?
(i) Which of the following equations corresponds to this number puzzle?

(a) $12n + 2 = 3n$ (b) $2n + 12 = 3$ (c) $2n + 12 = 3n$

(ii) Use the correct equation to solve the puzzle.
(iii) Check your solution works in the puzzle.

6. When 6 is added to a number, the result is 11.
Let x be the number.
(i) Write an equation for this sentence.
(ii) Solve the equation and check that your answer is correct.

7. When 4 is subtracted from twice a number, the result is 12.
Let x be the number.
(i) Write an equation for this sentence.
(ii) Solve the equation and check that your answer is correct.

8. When 4 times a number is increased by 3, the answer is 23.
Let x be the number.
(i) Write an equation for this sentence.
(ii) Solve the equation and check that your answer is correct.

9. When 6 is taken from five times a number, the answer is 14.
Find the number.

10. If I multiply a number by 3 and then subtract 4, the result is the same as twice the number. Find the number.

11. When 3 is taken from five times a number, the result is the same as adding 6 to twice the number. Find the number.

12. One number is 5 greater than another number.
(i) If we let x be the first number write an expression the second number.
(ii) If the two numbers are added together the result is 59. Find the numbers.